Products of Elliptic Indices and Finite Algebraic Correspondences

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Abstract

Let P generate a Zariski-dense cyclic subgroup of an abelian variety A. The triples $([n]P, [h]P, [nh]P)$ are Zariski dense in A^3, even if n ranges through an arbitrary infinite set and each n is assigned an arbitrary infinite set of values of h. For elliptic curves we give a direct proof by intersecting divisors with graphs of multiplication. For general abelian varieties the assertion follows from Mordell–Lang and a criterion in terms of affine relations among the indices. Consequently a fixed finite algebraic correspondence cannot reconstruct the index product from the two input points uniformly on these sets. This restriction does not apply to arbitrary Diophantine definitions or to unrestricted additional arithmetic conditions.

1 The density theorem

Throughout, k is a number field, all Zariski closures are geometric, and A/k is a positive-dimensional abelian variety. Let $P \in A(k)$, and suppose that $\Gamma = \mathbb{Z}P$ is Zariski dense in A. In particular P is not torsion.

Theorem 1.1. *Let $I \subseteq \mathbb{Z}$ be infinite. For each $n \in I$, let $J_n \subseteq \mathbb{Z}$ be infinite. Then*

$$\{([n]P, [h]P, [nh]P) : n \in I,\ h \in J_n\}$$

is Zariski dense in A^3.

The sets J_n may depend on n. In particular, arbitrary fixed congruence restrictions on the auxiliary index are allowed, provided an infinite set of indices remains for each n. When A is an elliptic curve, every nontorsion point satisfies the density hypothesis on Γ.

1.1 A direct proof for an elliptic curve

Proof of Theorem 1.1 when $A = E$. Suppose a proper closed subset $V \subsetneq E^3$ contains all the displayed triples. Let $\mathcal{L}$ be an ample line bundle of degree $d > 0$ on E, and put

$$\mathcal{H} = \mathcal{L} \boxtimes \mathcal{L} \boxtimes \mathcal{L}.$$

A projective embedding given by a sufficiently large power of $\mathcal{H}$, followed by a hypersurface containing V but not E^3, supplies a nonzero effective divisor

$$D \in |\mathcal{H}^{\otimes N}|, \qquad V \subseteq D,$$

for some fixed $N > 0$.

Only finitely many fibers of the first projection $E^3 \to E$ are contained in D: any such fiber is an irreducible divisor and hence an irreducible component of D. For every other $n \in I$, the

restriction D_n to the fiber over $[n]P$ is a nonzero effective divisor on E^2, with fixed numerical class

$$N(\mathcal{L} \boxtimes \mathcal{L}).$$

The points $([h]P, [nh]P)$, for $h \in J_n$, are dense in the graph

$$G_n = \{(Q, [n]Q) : Q \in E\},$$

because $J_n P$ is an infinite subset of the curve E. Thus G_n is an irreducible component of D_n.

Let $\mathcal{H}_0 = \mathcal{L} \boxtimes \mathcal{L}$ on E^2. Its self-intersection is $2d^2$, so

$$\deg_{\mathcal{H}_0}(D_n) = 2Nd^2.$$

On the other hand,

$$\deg_{\mathcal{H}_0}(G_n) = \deg(\mathcal{L} \otimes [n]^*\mathcal{L}) = d(1 + n^2),$$

using $\deg[n] = n^2$; see [3, Chapter I, Theorem 7.2]. Since $\mathcal{H}_0$ is ample and $D_n - G_n$ is effective,

$$d(1 + n^2) \leq 2Nd^2.$$

This is impossible for the unbounded set of integers $n \in I$ outside the finitely many excluded fibers. □

2 Affine relations among indices

We use the following form of the Mordell–Lang theorem: the intersection of a closed subvariety of an abelian variety with a finitely generated subgroup is a finite union of cosets of subgroups of that finitely generated group. This is a theorem, not an additional conjecture; see [1], and the precise formulation in the introduction of [2].

Proposition 2.1. *Let $r \geq 1$ and $\Sigma \subseteq \mathbb{Z}^r$. The set*

$$\{([m_1]P, \ldots, [m_r]P) : \mathbf{m} \in \Sigma\}$$

is Zariski dense in A^r if and only if Σ is not contained in a finite union of proper rational affine hyperplanes in $\mathbb{Q}^r$.

Proof. The product Γ^r is Zariski dense in A^r. Suppose the displayed set is contained in a proper closed subvariety V. By Mordell–Lang, $V \cap \Gamma^r$ is a finite union of cosets of subgroups of Γ^r. Identify Γ^r with $\mathbb{Z}^r$, using the nontorsion point P. None of these subgroups has rank r. Indeed a rank-r subgroup has finite index and contains $m\Gamma^r$ for some nonzero integer m; this is Zariski dense in A^r, as multiplication by m is surjective. Its translate cannot be contained in V.

Every subgroup of $\mathbb{Z}^r$ of rank at most $r - 1$ is annihilated by a nonzero integer linear form. Its cosets are therefore contained in proper rational affine hyperplanes. These finitely many hyperplanes contain Σ.

Conversely, after clearing denominators, an affine hyperplane has equation

$$a_1 m_1 + \cdots + a_r m_r = c, \qquad a_i, c \in \mathbb{Z},$$

with some $a_i \neq 0$. The corresponding points lie in the proper closed subset

$$[a_1]Q_1 + \cdots + [a_r]Q_r = [c]P$$

of A^r. This subset is proper because the indicated homomorphism $A^r \to A$ is surjective. A finite union of such proper closed subsets is still proper. □

Proof of Theorem 1.1 for general A. Apply Proposition 2.1 to the index triples (n, h, nh). On these triples, an affine hyperplane becomes

$$(b + cn)h + an = d, \qquad (a, b, c) \neq (0, 0, 0).$$

For all but at most one value of n, this equation has at most one solution h, or none. Indeed, if $b + cn \neq 0$, it determines h. If $b + cn = 0$ for all n, then $b = c = 0$, and $a \neq 0$ makes $an = d$ possible for at most one n. Thus finitely many such hyperplanes cannot contain infinitely many values of h for every n in the infinite set I. □

Corollary 2.2. *Let $f_1, \dots, f_r \in \mathbb{Z}[T_1, \dots, T_s]$, and suppose that $1, f_1, \dots, f_r$ are linearly independent over $\mathbb{Q}$. Then*

$$\{([f_1(\mathbf{t})]P, \dots, [f_r(\mathbf{t})]P) : \mathbf{t} \in \mathbb{Z}^s\}$$

is Zariski dense in A^r. In particular, for every polynomial $f \in \mathbb{Z}[T]$ of degree at least two and every infinite set $I \subseteq \mathbb{Z}$,

$$\{([n]P, [f(n)]P) : n \in I\}$$

is Zariski dense in A^2.

Proof. The preimage of a proper affine hyperplane is the zero set of a nonzero polynomial. A finite union of such zero sets cannot contain $\mathbb{Z}^s$, since their product is a nonzero polynomial. For the last statement, the nonzero polynomial $an + bf(n) - c$ has only finitely many roots. Now apply Proposition 2.1. □

3 The finite-correspondence consequence

Corollary 3.1. *There is no proper closed subset $C \subsetneq A^3$ containing all the triples in Theorem 1.1. In particular there is no fixed algebraic correspondence that has finite fibers over a nonempty open subset of A^2 and recovers the third coordinate $[nh]P$ from $([n]P, [h]P)$ for all these pairs in that open subset. The same assertion holds for a finite union of such correspondences, and for finite algebraic auxiliary choices followed by rational formulas for the third point.*

Proof. The first assertion is Theorem 1.1. For the next assertion, restrict to the nonempty open set where the correspondence has finite fibers. Each component dominating the base has dimension at most $2 \dim A$; its image in A^3 has proper closure. Components whose images do not dominate the base are contained over a proper closed subset of the base and also have proper image closure. The triples of Theorem 1.1 remain Zariski dense after restriction to the inverse image of a nonempty open set of A^2. They therefore cannot all lie in these finitely many proper closed images.

Finite algebraic auxiliary choices over the base give a generically finite variety over that base. Applying rational formulas and restricting to their domains reduces to the same dimension argument. □

Here finite fibers means zero-dimensional geometric fibers. Merely having finitely many rational points in a geometrically positive-dimensional fiber does not meet this hypothesis.

For an elliptic curve, this excludes replacing the point $U = [hn]P$ by any fixed finite algebraic choice constructed from $H = [h]P$ and $Q = [n]P$, while retaining infinitely many auxiliary indices for every member of an infinite set of labels. Corollary 2.2 similarly excludes a fixed finite algebraic construction of $[n^2]P$ from $[n]P$ on any infinite set of indices.

For a restriction of scalars, the general theorem applies once the chosen cyclic subgroup is known to be Zariski dense in that abelian variety. Its use does not follow from the cyclic

description of the rational points alone: the Zariski-density hypothesis must be checked in the relevant variety.

These are restrictions on the indicated reconstruction strategy. They do not exclude a source that retains only a specially chosen finite set of auxiliary indices for each input, changes the arithmetic conditions, uses a different algebraic group, introduces an additional free rational parameter, or defines the integers with fewer variables. The rational-root Pell strategy is not ruled out merely by these statements: its missing requirement is a complete global construction of the rational root, together with a sound argument controlling its denominators.

References

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[3] J. S. Milne, *Abelian Varieties*, version 2.0, 2008, https://www.jmilne.org/math/CourseNotes/AV.pdf.